

NETFLIX RECOMMENDATION SYSTEM

Personalized Content Discovery via Advanced Matrix Factorization

01 / THE MOTIVATION

The Era of Personalization

Every day, billions of users interact with recommendation systems across platforms like Netflix, Amazon, and Spotify.

Delivering relevant content directly influences **user engagement, retention**, and overall user experience.

Recommendation systems have evolved into the most critical component of modern digital streaming products.

The Core Challenge

Content overload causes **decision fatigue**. We need an intelligent engine to accurately predict user preferences.

Goal: Design a system to learn historical interactions, predict ratings for unseen content, and surface meaningful recommendations.

02 / DATASET OVERVIEW

Working with the legendary **Netflix Prize Dataset**—one of the most influential benchmark datasets in machine learning history.

100M+

MOVIE RATINGS

480K

UNIQUE USERS

17.7K

MOVIES

98.4%

MATRIX SPARSITY

The Sparsity Challenge

Because each user rates only a tiny fraction of available movies, building an engine requires overcoming extreme missing data, making it the perfect battleground for Latent Factor Modeling.

03 / EXPLORATORY DATA ANALYSIS

Key Behavioral Patterns

- **Long-Tail Activity:** A very small subset of highly active users generates the vast majority of all ratings.
- **Popularity Bias:** Rating volume is skewed heavily towards blockbuster hits, obscuring niche movies.
- **Sentiment Skew:** Average ratings skew positive (approx. 3.6 / 5.0), meaning users generally interact with content they already like.

Business Implication

Due to extreme sparsity and popularity bias, purely item-to-item correlation fails on niche movies.

A robust **matrix factorization** approach is strictly required to map users to less popular, highly relevant titles accurately.

04 / APPROACH & METHODOLOGY

We engineered and evaluated two distinct paradigms for capturing user preferences:

1. Item-Based Collaborative Filtering

Computes cosine similarity between items to find items similar to historical interactions.

- Highly interpretable models.
- Computationally expensive at scale.
- Highly vulnerable to the sparsity problem.

2. Singular Value Decomposition (SVD)

A powerful Matrix Factorization technique that reduces dimensionality to uncover **latent factors**.

- Discovers underlying movie traits and tastes.
- Superior handling of matrix sparsity.
- Exceptional predictive accuracy.

05 / EVALUATION STRATEGY

Rigorous Framework

Data was partitioned using a strict **80/20 Train-Test** methodology to validate generalized performance on unseen data.

Mandatory Metrics

- **RMSE**: Measures absolute rating prediction accuracy.
- **MAP@10**: Measures the quality of the Top-10 recommendation rankings.

Defining "Relevance"

For MAP@10 calculations, a movie is strictly considered **relevant** only if the actual ground-truth user rating is ≥ 3.5 .

The system generates Top-10 lists based on highest predicted scores, then checks against this relevance threshold to calculate Precision.

06 / EXPERIMENTAL RESULTS

Comparative Performance

The SVD architecture vastly outperformed traditional Collaborative Filtering in pure prediction accuracy.

SVD successfully minimized prediction error, proving its ability to extract meaningful latent structures from highly sparse data.

0.988

SVD RMSE ACHIEVED

Computational Efficiency vs. Quality

While SVD requires higher initial training complexity (calculating matrix gradients via stochastic gradient descent), the **inference time** is significantly faster and more accurate than computing neighborhood similarities on-the-fly.

07 / RECOMMENDATIONS & TAKEAWAYS

Top-K Generation Analysis

Success Cases: SVD successfully identifies niche sub-genres for users by matching latent factors, moving beyond simple popularity algorithms.

Failure Cases: Users with fewer than 5 ratings (The Cold-Start Problem) receive overly generalized recommendations due to lack of vector mapping.

Future Roadmap

- **Neural Collaborative Filtering:** Integrate Deep Learning to capture non-linear user-item relationships.
- **Hybrid Models:** Incorporate Movie Metadata (genres, cast) alongside explicit ratings to build a context-aware engine and solve cold-start issues.